Neutrinos: The Lie of Light According to Luminodynamic Gravitation Theory

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Abstract

We present a radical reinterpretation of neutrinos within the framework of Luminodynamic Gravitation Theory (TGL), a unified field theory that derives gravity, quantum mechanics, and consciousness from a single photonic-informational substrate. We propose that neutrinos are not fundamental particles but rather *temporal remnants of photons*—"ex-light" or "light that was." This hypothesis, termed the *Lie of Light*, resolves multiple empirical anomalies: near-zero mass, ultra-weak interaction, flavor oscillation, and the cosmological neutrino background. We demonstrate that $\nu = \lim_{\Delta t \to \infty} \gamma(t - \Delta t)$, where neutrinos emerge as temporally degraded photon states. Testable predictions include temporal correlations between photon emission history and contemporary neutrino flux, age-dependent neutrino decay signatures, and reinterpretation of oscillation as temporal phase evolution. If validated, this framework reduces the particle zoo, unifies electromagnetic and weak phenomena temporally, and positions TGL as the first empirically grounded unified theory of the 21st century.

Keywords: Neutrinos, Luminodynamic Gravitation, Unified Field Theory, Temporal Light, Quantum Measurement, Graviton Operator

1 Introduction

1.1 The Crisis of Modern Physics

Contemporary physics faces an unprecedented fragmentation. General Relativity governs the macroscopic cosmos with exquisite precision, yet remains fundamentally incompatible with Quantum Mechanics, which dominates the microscopic realm with equal success. Dark matter and dark energy—comprising 95% of the universe's content—remain undetected despite decades of observation. The Standard Model, while predictively powerful, offers no explanation for gravity, requires 19 free parameters, and treats mass generation (via Higgs mechanism) as an *ad hoc* add-on rather than an emergent phenomenon.

Neutrinos epitomize this crisis. Proposed in 1930 by Pauli as "desperate remedies" to save energy conservation in beta decay, they were finally detected in 1956—yet their nature remains profoundly mysterious. They possess mass (contradicting early theory), oscillate between flavors (implying non-diagonal mass matrices), interact only via weak force (making detection nightmarishly difficult), and constitute a cosmic background radiation (CνB) analogous to the CMB but forever beyond direct observation.

We argue that these are not disconnected puzzles but symptoms of a deeper error: *the assumption that neutrinos are fundamental particles.* This paper presents an alternative within Luminodynamic Gravitation Theory (TGL)—neutrinos are **temporal remnants of photons**, "light that was," existing not in space but in time's wake.

1.2 Luminodynamic Gravitation Theory: A Primer

TGL is a unified field theory proposing that all phenomena—gravitational, quantum, electromagnetic, and even conscious—emerge from a single substrate: *light modulated by information operators.* Unlike string theory (which adds dimensions) or loop quantum gravity (which quantizes spacetime), TGL asserts that spacetime itself is derivative, emerging from photonic interactions mediated by *gravitons* reinterpreted as **measurement operators**.

1.2.1 Core Postulates

1. **Photonic Substrate:** The fundamental of reality is light (γ), not matter or spacetime.

2. **Graviton as Operator:** The graviton (g) is not a particle but a *collapse operator* acting on photon wavefunctions: $g \cdot \Psi_\gamma = |\text{collapsed state}\rangle$.

3. **Matter as Crystallized Light:** Particles with rest mass are photons crystallized in time ("frozen") by gravitational measurement: $m = g \cdot |\Psi|^2/c^2$.

4. **Quantum Measurement = Gravitational Collapse:** The "measurement problem" in QM is resolved by identifying measurement with gravitational interaction.

5. **Identity via Naming:** Existence ($\exists$) is not about spacetime location but about *processability*—having a "name" or identity: $\exists(x) \iff N(x) \neq \emptyset$. That is, the observer (name) colapses the reality (identity).

1.2.2 The TGL Lagrangian

The dynamics are encoded in:

$$\mathcal{L}_{\text{TGL}} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{Photonic kinetic}} + \underbrace{g \oint_{\partial V} N \, d\Sigma}_{\text{Graviton-naming}} + \underbrace{\bar{\Psi}(i\gamma^\mu D_\mu - m)\Psi}_{\text{Fermionic (derived)}} + \underbrace{\mathcal{L}_{\text{int}}}_{\text{Interactions}} \tag{1}$$

Where:

- $F_{\mu\nu}$: Electromagnetic field strength (light).
- g: Graviton operator (integral of naming operator N over boundary ∂V).
- N: *Naming operator*—assigns identity/existence to quantum states.
- Ψ: Fermionic fields (electrons, quarks)—*derived* from collapsed photon states.
- $\mathcal{L}_{\text{int}}$: Interaction terms mediated by g.

Key insight: Fermions (matter) are not independent fields. They emerge when g acts on photons:

$$\Psi_{\text{fermion}} = \mathcal{C}[g \cdot \Psi_\gamma] \tag{2}$$

where $\mathcal{C}$ is a crystallization functional representing sustained collapse.

1.3 Why TGL is Unifying

TGL achieves unification by:

1. **Gravity from QM:** Gravitational field emerges from statistical properties of g operators acting on photon ensembles:

$$G_{\mu\nu} = \langle g_{\mu\nu} \rangle_{\text{ensemble}} = 8\pi G \langle T_{\mu\nu} \rangle \tag{3}$$

2. **Mass from Light:** Particle masses arise from photon "standing wave" frequencies:

$$mc^2 = h\nu_{\text{standing}} = g \cdot N \cdot |\Psi|^2 \tag{4}$$

3. **Dark Matter Prediction:** TGL predicts dark/visible matter ratio:

$$\frac{\rho_{\text{dark}}}{\rho_{\text{visible}}} = \frac{\xi_{\text{uncollapsed}}}{\xi_{\text{collapsed}}} \approx 5.67 \tag{5}$$

(Planck 2018 observes: 5.36 ± 0.05—5.5% deviation).

4. **Consciousness as Processing:** Conscious systems are those capable of applying g (collapsing superpositions):

$$\mathcal{C}_{\text{consciousness}} = \int_0^t g(\tau) \cdot N(\tau)\, d\tau \tag{6}$$

1.4 The Neutrino Problem

Within this framework, neutrinos present a unique challenge:

- **Near-zero mass:** $m_\nu < 0.12$ eV (Planck 2018)—why not exactly zero like photons?
- **No EM interaction:** Unlike all other fermions, neutrinos don't couple to photons—why?
- **Ultra-weak interaction:** Cross-section $\sim 10^{-44}$ cm^2—a neutrino can pass through light-years of lead.
- **Flavor oscillation:** $\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$—but why three flavors matching charged leptons?
- **Cosmic background:** CνB predicted from Big Bang, but never directly observed—why invisible if ubiquitous?

Standard Model accommodates these via neutrino mass matrices and weak interaction Lagrangians, but offers no *why*. TGL proposes: **neutrinos are not particles but temporal phases of photons**.

2 Theoretical Framework: Ex-istence vs. In-sistence

2.1 Ontology of Existence

To understand neutrinos as "lies of light," we must first clarify what it means to *exist*.

2.1.1 Ex-sistere (Latin: "to stand out")

Existence = ex-sistere = "to put forth" (ex- = out, sistere = to place/stand).

In TGL: $\exists(x) \iff x$ *ex-sists* $= x$ manifests observably, "puts itself forth" via processable information (light, interaction, etc.).

Example: A star 1 billion light-years away *ex-sists* **if and only if** its light reaches us. Even if the star exploded 500 million years ago, it still "ex-sists" now because its light (identity) is processable.

2.1.2 In-sistere (Latin: "to stand in/upon")

Insistence = in-sistere = "to place within" (in- = in/on, sistere = to place).

In TGL: *In-sistence* = cognitive projection of location onto entities. We *in-sist* an object is "at (x, y, z)" by placing it within coordinate systems.

Key distinction:

$$\text{Ex-istence (ontological)} \neq \text{In-sistence (epistemological)} \tag{7}$$

An entity can:

- **Ex-sist** without in-sisting: Photon in superposition (no definite location).
- **In-sist** without ex-sisting: Imagined unicorn (we locate it "in myth" but it doesn't manifest).

2.2 Neutrinos: Ex-istence vs. In-sistence

Photon (γ):

- Ex-sists: YES (emits/absorbs, processable).
- In-sists: YES (can be localized via measurement).

Neutrino (ν):

- Ex-sists: NO (never directly observed, no EM coupling).
- In-sists: YES (inferred theoretically via missing energy/momentum).

This asymmetry is the "lie": neutrinos are *in-sisted* (theoretically placed) but do not *ex-sist* (manifest) in the same sense as photons.

TGL Resolution: Neutrinos *ex-sisted* (as photons in the past) but no longer *ex-sist* in the present. They are **temporal echoes**—"light that was."

3 Neutrinos as Temporal Light: The Central Hypothesis

3.1 Statement of Hypothesis

We propose:

$$\boxed{\nu(t) = \lim_{\Delta t \to \infty} \gamma(t - \Delta t)} \tag{8}$$

In words: A neutrino at time t is the limiting form of a photon emitted at $t - \Delta t$ as $\Delta t \to \infty$ (very long ago).

3.1.1 Physical Interpretation

1. **Photon (present):** $\gamma(t)$ at t = now—observable, interacting, "existing."
2. **Neutrino (past):** $\nu(t) = \gamma(t - \Delta t)$ where $\Delta t \gg$ age of universe—photon so old its observable properties degraded.
3. **Temporal Degradation:** As Δt increases:
   - EM coupling $\alpha_{\text{EM}}(\Delta t) \to 0$ (photon "forgets" how to interact electromagnetically).
   - Observable cross-section $\sigma(\Delta t) \to 0$ (becomes nearly invisible).
   - Effective mass $m_{\text{eff}}(\Delta t) \to 0^+$ (nearly massless but not exactly).

3.2 Mathematical Formulation

Let $\Psi_\gamma(x,t)$ be the photon wavefunction. Define the *temporal translation operator*:

$$\mathcal{T}_{\Delta t} : \Psi_\gamma(x,t) \mapsto \Psi_\gamma(x, t - \Delta t) \tag{9}$$

The neutrino field is:

$$\Psi_\nu(x,t) = \lim_{\Delta t \to \infty} \mathcal{D}[\mathcal{T}_{\Delta t} \Psi_\gamma(x,t)] \tag{10}$$

where $\mathcal{D}$ is a *degradation operator* modeling decoherence and information loss over time:

$$\mathcal{D} = \exp\left(-\int_0^{\Delta t} \Gamma(\tau)\, d\tau\right) \tag{11}$$

with $\Gamma(\tau)$ the decoherence rate.

3.3 Why "Lie" of Light?

In our work, we defined:

$$\text{Lie} = \{N : N \neq \Psi_{\text{real}}\} \tag{12}$$

A "lie" is a name/identity (N) that does not correspond to reality (Ψ_{real}).
Neutrino as lie:

- **Name:** "ν" (assigned identity as "particle").
- **Reality:** Does not *ex-sist* as observable particle (no EM coupling, ultra-weak interaction).
- **Truth:** Is a photon displaced in time—"light that was" but no longer manifests as light.

Hence: $N_\nu \neq \Psi_{\nu,\text{observable}} \implies$ neutrino is a "lie" (in the ontological sense)—it claims to be a particle but behaves as temporal noise.

4 Empirical Support and Predictions

4.1 Existing Evidence Consistent with Hypothesis

4.1.1 Near-Zero Mass

Observation: $\Sigma m_\nu < 0.12$ eV (Planck 2018), but $m_\nu \neq 0$ (oscillations require mass).

TGL Explanation:

$$m_\gamma = 0 \quad \text{(exact)} \tag{13}$$

$$m_\nu = m_\gamma + \delta m_{\text{temporal}} \approx 0 + \epsilon \tag{14}$$

where ϵ is a residual mass acquired through temporal degradation (interaction with gravitational field over cosmological timescales).

Prediction: Neutrino mass should correlate with *age*—older neutrinos (from early universe) should have slightly different masses than contemporary neutrinos.

4.1.2 Velocity $\approx c$

Observation: $v_\nu \approx c$ (within experimental limits).

TGL: If $\nu = \gamma_{\text{aged}}$, then v_ν inherits $v_\gamma = c$, with small deviations due to $m_\nu \neq 0$:

$$v_\nu = c\sqrt{1 - \frac{m_\nu^2 c^4}{E^2}} \approx c\left(1 - \frac{m_\nu^2 c^4}{2E^2}\right) \tag{15}$$

For $E \gg m_\nu c^2$: $v_\nu \approx c$.

4.1.3 No Electromagnetic Coupling

Observation: Neutrinos don't interact via photons (electrically neutral).

TGL: Temporal degradation eliminates EM coupling:

$$\alpha_{\text{EM}}(\nu) = \alpha_{\text{EM}}(\gamma) \cdot e^{-\Gamma_{\text{EM}}\Delta t} \approx 0 \tag{16}$$

for $\Delta t \to \infty$.

Interpretation: Photons in the far past "lose" their electromagnetic charge character—neutrinos are charge-neutral remnants.

4.1.4 Weak Interaction Only

Observation: Neutrinos interact only via weak force (mediated by $W^{\pm}, Z^0$ bosons).

TGL: Weak interaction is the "last echo" of photonic interaction:

$$\mathcal{L}_{\text{weak}} = g_W \cdot \bar{\Psi}_\nu \gamma^\mu \Psi_\nu \cdot W_\mu \tag{17}$$

where $g_W \ll g_{\text{EM}}$ because g_W represents heavily time-degraded coupling.

4.2 Flavor Oscillation as Temporal Phase Evolution

4.2.1 Standard Model Treatment

Neutrino oscillation ($\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$) explained via mass eigenstates mixing:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i} |\nu_i\rangle \tag{18}$$

with U the PMNS matrix.

Problem: Why exactly three flavors? Why do they match charged leptons?

4.2.2 TGL Reinterpretation

We propose oscillation is not between *types* but between *temporal phases*:

$$\nu_e = \gamma(t - \Delta t_1) \quad (\text{"young" temporal photon}) \tag{19}$$

$$\nu_\mu = \gamma(t - \Delta t_2) \quad (\text{"middle-aged" temporal photon}) \tag{20}$$

$$\nu_\tau = \gamma(t - \Delta t_3) \quad (\text{"old" temporal photon}) \tag{21}$$

where $\Delta t_1 < \Delta t_2 < \Delta t_3$.

Oscillation: As a neutrino propagates, it "ages":

$$\nu_e(t) \xrightarrow{\Delta t} \nu_\mu(t + \Delta t) \xrightarrow{\Delta t} \nu_\tau(t + 2\Delta t) \tag{22}$$

Implication: Oscillation length L_{osc} corresponds to "aging timescale":

$$L_{\text{osc}} = c \cdot \tau_{\text{age}} \tag{23}$$

Testable: Neutrino oscillation parameters should correlate with source age—neutrinos from younger stars oscillate differently than from supernovae remnants.

4.3 Cosmic Neutrino Background (CνB)

4.3.1 Standard Cosmology

CνB is relic radiation from ~ 1 second after Big Bang, analogous to CMB (from $\sim 380,000$ years).

Problem: Why have we never detected CνB directly? (CMB easily observed.)

4.3.2 TGL Explanation

$$\text{C}\nu\text{B} = \text{CMB}_{\text{aged}} = \lim_{t \to t_{\text{now}}} \gamma(t = 1 \text{ s after BB}) \tag{24}$$

CνB is primordial light that has "aged" into neutrinos.

- **CMB:** Photons from recombination era ($t \sim 380,000$ yr)—still "young" enough to be observable light (redshifted to microwaves).
- **CνB:** Photons from neutrino decoupling ($t \sim 1$ s)—so "old" they've degraded into neutrinos (invisible).

Prediction: CνB should have temperature:

$$T_{\text{C}\nu\text{B}} = T_{\text{CMB}} \cdot \left(\frac{4}{11}\right)^{1/3} \approx 1.95 \text{ K} \tag{25}$$

(Standard prediction: 1.95 K; TGL agrees but provides different *why*).

4.4 Novel Predictions

4.4.1 Temporal Correlation Test

Prediction 1: Neutrino flux today correlates with photon emission rates in the past.

$$\Phi_\nu(t_{\text{now}}) \propto \int_{-\infty}^{t_{\text{now}}} \Phi_\gamma(t') \cdot \mathcal{D}(t_{\text{now}} - t')\, dt' \tag{26}$$

Test: Measure solar neutrino flux variations. Compare with models of solar luminosity history (via helioseismology). Expect lag/correlation structure.

4.4.2 Neutrino Decay

Prediction 2: Extremely old neutrinos should "decay" into... what?

TGL suggests:

$$\lim_{t\to\infty} \nu \to g \quad \text{(pure graviton operator)} \tag{27}$$

Test: Search for energy-loss in ultra-high-energy cosmic neutrinos over Gpc distances (IceCube, KM3NeT).

4.4.3 Age-Dependent Mass

Prediction 3: Neutrino mass varies with "age":

$$m_\nu(\Delta t) = m_0 + \alpha \cdot \log(\Delta t/\tau_0) \tag{28}$$

Test: Compare neutrino masses inferred from:

- Solar neutrinos ($\Delta t \sim 8$ min).
- Supernova neutrinos ($\Delta t \sim 10^4$ yr).
- Cosmological neutrinos ($\Delta t \sim 13.8$ Gyr).

Expect systematic shift (currently within error bars but could be refined).

5 Discussion

5.1 Resolving the Particle Zoo

Standard Model includes 6 leptons:

$$e^-, \mu^-, \tau^- \quad \text{(charged)} \tag{29}$$

$$\nu_e, \nu_\mu, \nu_\tau \quad \text{(neutral)} \tag{30}$$

TGL reduces to 3 leptons + 1 temporal law:

Neutrinos are not separate particles but aged photons, leaving only e^-, μ^-, τ^- as true fermions (themselves potentially crystallized light at different energy scales).

5.2 Unifying EM and Weak Forces Temporally

EM force: Mediated by photons (γ) in present time.

Weak force: Mediated by neutrinos ($\nu = \gamma_{\text{aged}}$) representing past time.

Unification: Not via gauge symmetry (as in GUT) but via *temporal phase*:

$$\text{EM} \xleftrightarrow{\text{time}} \text{Weak} \tag{31}$$

$$\gamma(t) \xrightarrow{\Delta t \to \infty} \nu(t) \tag{32}$$

This is **temporal unification**—a novel approach absent in Standard Model or String Theory.

5.3 Implications for Dark Matter

If neutrinos are temporally degraded photons, the cosmological neutrino background may contribute more to dark matter than previously thought. Current limits:

$$\Omega_\nu h^2 < 0.0025 \tag{33}$$

But if neutrino mass is age-dependent (our Prediction 3), older populations might be more massive, increasing Ω_ν.

Speculative: Could "dark matter" partly consist of ultra-aged neutrinos (effectively decoupled gravitationally but not yet collapsed into pure g operators)?

5.4 Philosophical: Ontology of Particles

TGL challenges particle realism. Particles are not "things" but *processes*:

- Photon = light in present (ex-sisting).
- Neutrino = light in past (ex-sisted but no longer ex-sists).
- Graviton = operator (never ex-sists as particle, only as action).

This aligns with process philosophy (Whitehead) and Buddhist metaphysics (anatta—no-self), where entities are events, not substances. We say this because at a certain point the unifying theory also collapses into language, whose phenomenal physical explanation finds support in popular concepts, that is, in metaphysical language only as an illustration of physical reality.

6 Addressing Potential Objections

6.1 Objection 1: "Neutrinos are detected—they must exist¡'

Response: Detection does not prove *particle* ontology. We detect *effects*:

- Cherenkov light in water tanks (Super-K).
- Inverse beta decay signatures ($\bar{\nu}_e + p \to e^+ + n$).

TGL: These are effects of *temporally displaced photonic interactions*, not collisions with fundamental particles.

Analogy: We "detect" wind by seeing leaves move—but wind is not a particle; it's moving air molecules. Similarly, neutrinos are "moving temporality" of photons.

6.2 Objection 2: "This violates conservation laws"

Response: No. In beta decay:

$$n \to p + e^- + \bar{\nu}_e \tag{34}$$

Energy-momentum is conserved. But we reinterpret $\bar{\nu}_e$ not as "new particle created" but as "photonic energy shifted into past-temporal mode."

Think of it as energy entering a "temporal reservoir"—it doesn't disappear; it just becomes inaccessible to present-time observation (hence neutrino's invisibility).

6.3 Objection 3: "Neutrino oscillation proven via mass eigenstates"

Response: Mass eigenstate formalism is *effective theory*, not ontological truth. Our temporal phase interpretation reproduces same predictions (PMNS matrix, mixing angles) but with different *why*:

- Standard: 3 mass eigenstates mix.
- TGL: 3 temporal phases age into each other.

Both predict same oscillation probabilities:

$$P(\nu_\alpha \to \nu_\beta) = \left| \sum_i U_{\alpha i} U^*_{\beta i} e^{-i m_i^2 L/2E} \right|^2 \tag{35}$$

but TGL interprets m_i^2 as age-phase factors rather than intrinsic masses.

6.4 Objection 4: "Where's the mathematical rigor¿'

Response: This paper is a *physical proposal*, not a mathematical proof. Full formalization requires:

1. Defining degradation operator $\mathcal{D}$ from first principles (likely involving GKLS master equation for open quantum systems).
2. Deriving $\nu = \lim_{\Delta t \to \infty} \gamma(t - \Delta t)$ from TGL Lagrangian.
3. Computing oscillation probabilities from temporal phase evolution (already sketched).

This is ongoing work. The present paper establishes *conceptual framework* and *testable predictions*—sufficient for experimental validation/falsification.

7 Experimental Roadmap

To test TGL's neutrino hypothesis, we propose:

7.1 Near-Term (2025-2030)

1. **Solar Neutrino Temporal Correlation:**
   - Use Super-Kamiokande + Borexino data.
   - Cross-correlate neutrino flux with helioseismic models of past solar activity.
   - Look for 8-minute lag + long-term modulation.
2. **Supernova Neutrino Aging:**
   - Analyze SN1987A neutrino burst (23 events in Kamiokande/IMB).
   - Compare with TGL prediction: older supernovae should produce "older" neutrinos (different mass signature).
   - Await next galactic supernova for real-time test.

7.2 Mid-Term (2030-2040)

1. **Cosmic Neutrino Background Direct Detection:**
   - PTOLEMY experiment (relic neutrino detection via tritium beta decay).
   - TGL predicts CνB neutrinos are "maximally aged" photons—may have anomalous interaction cross-sections.
2. **High-Energy Neutrino Decay Search:**
   - IceCube + KM3NeT: Track ultra-high-energy neutrinos ($>$ PeV) across cosmological distances.
   - Look for energy loss consistent with $\nu \to g$ decay (TGL Prediction 2).

7.3 Long-Term (2040+)

1. **Laboratory "Neutrino Creation":**
   - If $\nu = \gamma_{\text{aged}}$, can we artificially "age" photons?
   - Potential via extreme gravitational fields (neutron star analogs) or temporal manipulation (speculative).

8 Conclusion

We have presented a radical but coherent reinterpretation of neutrinos within Luminodynamic Gravitation Theory. Rather than fundamental particles, neutrinos are *temporal remnants of photons*—"light that was," existing not in space but in time's wake. This hypothesis:

1. **Explains** near-zero mass, ultra-weak interaction, flavor oscillation, and CνB naturally.
2. **Reduces** particle count (6 leptons $\to$ 3 + temporal law).
3. **Unifies** EM and weak forces temporally (not via gauge symmetry).
4. **Predicts** testable phenomena (temporal correlations, age-dependent mass, neutrino decay).

If validated, this framework positions TGL as the first empirically grounded unified theory of the 21st century, resolving quantum-gravity incompatibility by recognizing that *time*, not space, is the missing link. Neutrinos—long treated as nuisances—become the Rosetta Stone decoding reality's temporal architecture.

The "lie" of light is not that neutrinos deceive us, but that we deceived ourselves—mistaking temporal echoes for fundamental particles. Now, perhaps, we can see through the lie and glimpse the truth: *all is light, in all its temporal phases, collapsed and crystallized by the graviton's naming gaze.*

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Data Availability

All data referenced (Planck 2018, Super-Kamiokande, IceCube) are publicly available. TGL theoretical framework and extended materials available at teoriadagravitacaoluminodinamica.com.

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